

# Module 14: Error Handling and Debugging

## 14.1 Types of Errors

- Syntax errors: violations of C++ grammar caught by the compiler before the program runs
- Runtime errors: occur while the program is running (division by zero, null pointer dereference, array out of bounds)
- Logic errors: the program runs without crashing but produces incorrect output (wrong formula, wrong condition)

## 14.2 Debugging Strategies

- Read the compiler error message carefully: it tells you the exact line number and what is wrong
- Use cout statements to print variable values at different points to trace what is happening
- Use the Visual Studio debugger
  - Set a breakpoint by clicking the grey margin to the left of the line number (a red dot appears)
  - Press F5 to run in debug mode; execution pauses at breakpoints
  - Hover over a variable while paused to see its current value
  - Use Step Over (F10) to execute one line at a time
  - Use Step Into (F11) to enter a function and debug inside it
- Test with boundary values: zero, negative numbers, very large numbers, empty strings
- Check for off-by-one errors in loops (using < vs <= by mistake)

## 14.3 Exception Handling

- An exception is an error that occurs at runtime and disrupts the normal program flow
- Exception handling using try, throw, and catch

```
try {  
    if (denominator == 0) throw "Division by zero!";  
    int result = numerator / denominator;  
    cout << result;  
} catch (const char *msg) {  
    cout << "Error: " << msg;  
}
```

- Multiple catch blocks can handle different exception types
- The standard exception class (std::exception) and its subclasses